

A Technical Introduction to AlphaGo

Section 1

AI community AlphaGo's March 2016 victory was a major milestone in artificial intelligence research. Go had previously been regarded as a hard problem in machine learning that was expected to be out of reach for the technology of the time. Most experts thought a Go program as powerful as AlphaGo was at least five years away; some experts thought that it would take at least another decade before computers would beat Go champions. Most observers at the beginning of the 2016 matches expected Lee to beat AlphaGo. With games such as checkers (that has been solved by the Chinook computer engine), chess, and now Go won by computers, victories at popular board games can no longer serve as major milestones for artificial intelligence in the way that they used to. Deep Blue's Murray Campbell called AlphaGo's victory "the end of an era... board games are more or less done and it's time to move on." When compared with Deep Blue or Watson, AlphaGo's underlying algorithms are potentially more general-purpose and may be evidence that the scientific community is making progress towards artificial general intelligence. Some commentators believe AlphaGo's victory makes for a good opportunity for society to start preparing for the possible future impact of machines with general purpose intelligence. As noted by entrepreneur Guy Suter, AlphaGo only knows how to play Go and doesn't possess general-purpose intelligence; "[It] couldn't just wake up one morning and decide it wants to learn how to use firearms." AI researcher Stuart Russell said that AI systems such as AlphaGo have progressed quicker and become more powerful than expected, and we must therefore develop methods to ensure they "remain under human control". Some scholars, such as Stephen Hawking, warned (in May 2015 before the matches) that some future self-improving AI could gain actual general intelligence, leading to an unexpected AI takeover; other scholars disagree: AI expert Jean-Gabriel Ganascia believes that "Things like 'common sense'... may never be reproducible", and says "I don't see why we would speak about fears. On the contrary, this raises hopes in many domains such as health and space exploration." Computer scientist Richard Sutton said "I don't think people should be scared... but I do think people should be paying attention." In China, AlphaGo was a "Sputnik moment" which helped convince the Chinese government to prioritize and dramatically increase funding for artificial intelligence. In 2017, the DeepMind AlphaGo team received the inaugural IJCAI Marvin Minsky medal for Outstanding Achievements in AI. "AlphaGo is a wonderful achievement, and a perfect example of what the Minsky Medal was initiated to recognise", said Professor Michael Wooldridge, Chair of the IJCAI Awards Committee. "What particularly impressed IJCAI was that AlphaGo achieves

what it does through a brilliant combination of classic AI techniques as well as the state-of-the-art machine learning techniques that DeepMind is so closely associated with. It's a breathtaking demonstration of contemporary AI, and we are delighted to be able to recognise it with this award."

Section 2

Versions An early version of AlphaGo was tested on hardware with various numbers of CPUs and GPUs, running in asynchronous or distributed mode. Two seconds of thinking time was given to each move. The resulting Elo ratings are listed below. In the matches with more time per move higher ratings are achieved.

Section 3

Technology and AI-related fields James Vincent, a reporter from The Verge, comments that "It prods and pokes viewers with unobvious emotional cues, like a reality TV show would. "Now, you should be nervous; now you should feel relieved". The AlphaGo footage slowly captures the moment when Lee Sedol acknowledges the true power of AlphaGo AI. In the first game, he had more experience than his human-programmed AI, so he thought it would be easy to beat the AI. However, the early game dynamics were not what he expected. After losing the first match, he became more nervous and lost confidence. Afterward, he reacted to attacks by saying that he just wanted to win the match, unintentionally displaying his anger, and acting in an unusual way. Also, he spends 12 minutes on one move, while AlphaGo only takes a minute and a half to respond. AlphaGo weighs each alternative equally and consistently. No reaction to Lee's fight. Instead, the game continues as if he was not there. James also said that "suffice to say that humanity does land at least one blow on the machines, through Lee's so-called "divine move". "More likely, the forces of automation we'll face will be impersonal and incomprehensible. They'll come in the form of star ratings we can't object to, and algorithms we can't fully understand. Dealing with the problems of AI will take a perspective that looks beyond individual battles. AlphaGo is worth seeing because it raises these questions" (Vincent, 2017) Murray Shanahan, a professor of cognitive robotics at Imperial College London, critics that "Go is an extraordinary game but it represents what we can do with AI in all kinds of other spheres," says Murray Shanahan, professor of cognitive robotics at Imperial College London and senior research scientist at DeepMind, says. "In just the same way there are all kinds of realms of possibility within Go that have not been discovered, we could never have imagined the potential for discovering drugs and other materials."

Section 4

AlphaGo's team published an article in the journal Nature on 19 October 2017, introducing AlphaGo Zero, a version without human data and stronger than any previous

human-champion-defeating version. By playing games against itself, AlphaGo Zero surpassed the strength of AlphaGo Lee in three days by winning 100 games to 0, reached the level of AlphaGo Master in 21 days, and exceeded all the old versions in 40 days. In a paper released on arXiv on 5 December 2017, DeepMind claimed that it generalized AlphaGo Zero's approach into a single AlphaZero algorithm, which achieved within 24 hours a superhuman level of play in the games of chess, shogi, and Go by defeating world-champion programs, Stockfish, Elmo, and 3-day version of AlphaGo Zero in each case.

Section 5

Go community Go is a popular game in China, Japan and Korea, and the 2016 matches were watched by perhaps a hundred million people worldwide. Many top Go players characterized AlphaGo's unorthodox plays as seemingly-questionable moves that initially befuddled onlookers, but made sense in hindsight: "All but the very best Go players craft their style by imitating top players. AlphaGo seems to have totally original moves it creates itself." AlphaGo appeared to have unexpectedly become much stronger, even when compared with its October 2015 match where a computer had beaten a Go professional for the first time ever without the advantage of a handicap. The day after Lee's first defeat, Jeong Ahram, the lead Go correspondent for one of South Korea's biggest daily newspapers, said "Last night was very gloomy... Many people drank alcohol." The Korea Baduk Association, the organization that oversees Go professionals in South Korea, awarded AlphaGo an honorary 9-dan title for exhibiting creative skills and pushing forward the game's progress. China's Ke Jie, an 18-year-old generally recognized as the world's best Go player at the time, initially claimed that he would be able to beat AlphaGo, but declined to play against it for fear that it would "copy my style". As the matches progressed, Ke Jie went back and forth, stating that "it is highly likely that I (could) lose" after analysing the first three matches, but regaining confidence after AlphaGo displayed flaws in the fourth match. Toby Manning, the referee of AlphaGo's match against Fan Hui, and Hajin Lee, secretary general of the International Go Federation, both reason that in the future, Go players will get help from computers to learn what they have done wrong in games and improve their skills. After game two, Lee said he felt "speechless": "From the very beginning of the match, I could never manage an upper hand for one single move. It was AlphaGo's total victory." Lee apologized for his losses, stating after game three that "I misjudged the capabilities of AlphaGo and felt powerless." He emphasized that the defeat was "Lee Se-dol's defeat" and "not a defeat of mankind". Lee said his eventual loss to a machine was "inevitable" but stated that "robots will never understand the beauty of the game the same way that we humans do." Lee called his game four victory a "priceless win that I (would) not exchange for anything."